Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/10/2007
PI Name:	Kuznetsov, Andrey Ph.D.		
Project Title:	Investigation of Interactions between Bioconvection and Natural Convection and Biofilm Growth in Porous Media		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-7910	**Congressional District:**	4
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Task Description:	The conducted research dealt with one of the most vibrant areas of modern fluid dynamics, namely, microbiological fluid mechanics. Different from solid particles in traditional multiphase flow systems, many types of bacteria in bacterial suspension are self-propelled. They tend to swim in a particular direction in response to certain stimuli, such as gravity (gravitaxis), light (phototaxis), or chemical gradients (chemotaxis). Bacteria propel themselves by rotating flagella that are driven by reversible molecular motors that are embedded in the cell wall. In many cases the molecular motor periodically reverses its direction of rotation, producing on average irregular locomotion toward some specific direction. The number of these self-propelled bacteria in one cubic centimeter of suspension is usually very large (about 107 for a dilute regime up to about 1011 for a close packed (turbulent) regime), which results in very complex flow patterns. The conducted research project was divided into two parts. The first part dealt with the development of a continuum model of bioconvection in a porous medium. This was a novel project because continuum model of bioconvection was developed before only for microorganisms suspended in a clear fluid. Another novel feature of this project was the development of a model that described the interaction between bioconvection and natural convection in a porous medium. This investigation resulted in a discovery of new instabilities, which did not occur when bioconvection only or natural convection only was studied. The developed model included diffusion terms to account for the random element in bacterial motion. Accounting for randomness shifted the stability criteria; therefore the stability analysis was repeated after the diffusion terms modeling the stochastic behavior of microorganisms were incorporated into the governing equations. The second part dealt with direct simulation of bacterial motion in porous media and influence of bacterial motility and bioconvection on biofilm formation in porous rocks. This was also a novel project because nobody thus far tried to simulate directly bacterial swimming in two-dimensional configurations, as well as investigate such effects as collisions between two microorganisms and a change in the direction of swimming of a bacterium when it enters a propulsive stream caused by another bacterium. The PI also investigated interactions between bacteria and suspended solid particles. This means that he investigated the stability of a suspension that included two types of particles: (1) active, self-propelled microorganisms, and (2) passive solid particles. He investigated how bacterial and particle size distributions influenced bacterial motion and biofilm formation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has a broad an impact on biotechnology. This is because this research suggests a method of mixing micro volumes of fluid. This will have a positive impact on pharmaceuticals, public health, laboratory testing, agriculture, and other areas where the ability to move, pump, mix, and otherwise manipulate micro volumes of fluids is important. In addition to this, this research further advances the area of microbiological fluid mechanics, which is one of the most vibrant areas of modern fluid dynamics.
Task Progress:	The objectives for this study were: (1) Development of a continuum model of bioconvection in a porous medium. Investigation of the development of Rayleigh-Taylor instability During this phase of the project, the PI developed a model of bioconvection in porous media and analyzed the development of Raleigh-Taylor plumes. The results obtained by the PI indicated that the development of this instability strongly depended on permeability of the porous medium (if permeability was very low, the instability did not develop). The PI carried out a linear stability analysis of this problem and determined the critical value of permeability when the instability appeared. He also solved this problem numerically and investigated the development of gydrotactic plumes in bioconvection in a porous medium. (2) Extending the continuum model to investigate the interaction between bioconvection and natural convection in a porous medium At this stage, the model was supplemented with the energy equation to investigate an interaction between bioconvection and natural convection. This investigation resulted in a discovery of new types of instabilities. Next, the model was supplemented with the diffusion terms to account for the random element in bacterial motion. Accounting for randomness shifted the stability criteria; therefore the stability analysis was repeated after diffusion terms were introduced in the equations. (3) Investigation of the effect of bioconvection on number density distribution of dead cells in a suspension of gyrotactic motile microorganisms that contained both living and dead cells. This investigation was important to understand to what degree bioconvection complicated the separation between living and dead cells (modeled as inert particles). With no bioconvection, living cells concentrated in the upper part of the fluid layer and dead cells concentrated in the lower part of the fluid layer. Even if there were no convective motion of the fluid, the separation between the living and dead cells was not perfect. The difficulty resulted from the random component in the motion of microorganisms (which was modeled by cell diffusion). In the case of no bioconvection the distribution of the number density of living cells was determined by the balance between the upswimming of the cells and diffusion. For the same case, the distribution of the number density of dead cells was determined by the balance of buoyancy (that determined the sinking velocity of the dead cells) and diffusion. For the case of no bioconvection the maximum number density of the living cells occurred at the free surface while the maximum number density of the dead cells occurred at the bottom of the fluid layer. If bioconvection was present, the living and dead cells were considerably more mixed up. (4) Investigation of stability of suspensions of gyrotactic motile microorganisms in porous media and analyzing the effect of positioning porous plugs on suppressing bioconvection. This task included investigation of stability of a suspension of gyrotactic motile microorganisms in a fluid layer as well as the solution of the optimization problem of positioning porous plugs in a clear fluid domain to suppress bioconvection and optimize the separation between living and dead cells. (5) Development of a model of biofilm formation in porous media During this phase of the project, a model of biofilm accumulation was developed. Initially, the model was developed

	assuming that that all bacteria have a spherical shape, same radius, same density, and motility characteristics (such as the average velocity during the run intervals, and duration of the straight swimming and tumbling). Also, all solid particles were assumed spherical having the same radius. Computations were performed for both the dilute regime when collisions between bacteria were negligible and also for the denser regime when some bacteria collided or one bacterium entered a propulsion flow of another bacterium.
Bibliography Type:	Description: (Last Updated: 04/27/2010)
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